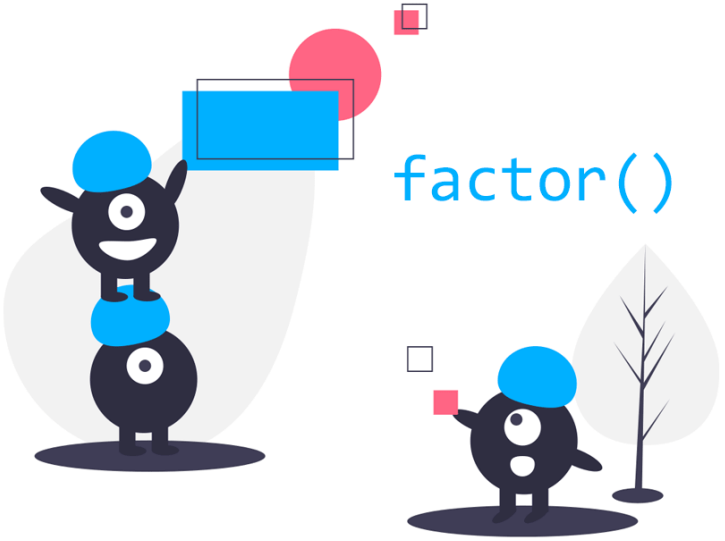




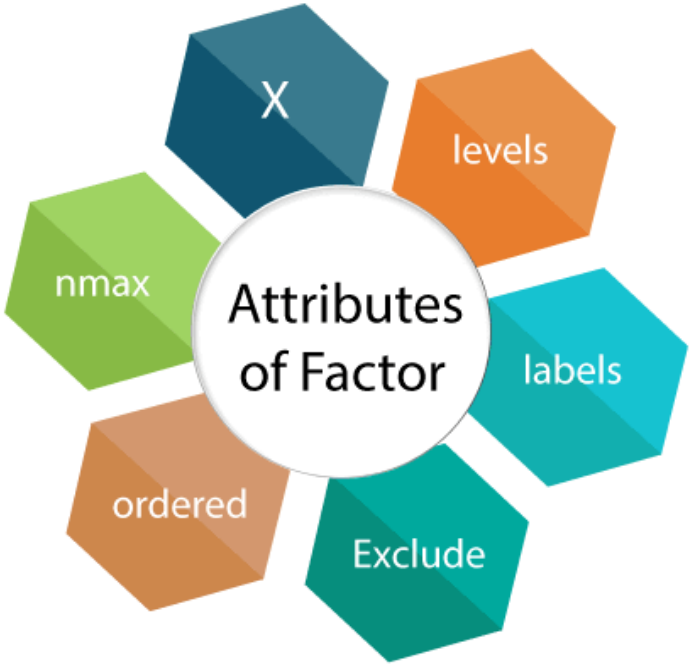
INSP/ESPM
Escuela de Salud Pública de México

R {forcats}, {lubridate}

José Luis Texcalac Sangrador

Procesamiento, visualización y modelado de datos espaciales en R

Factores en R



- Se usan para trabajar con variables categóricas, es decir, variables que tienen un conjunto fijo y conocido de valores posibles.
- Son útiles cuando quieres mostrar vectores de caracteres en un orden no alfabético.
- Los valores que un factor puede contener están delimitados por los niveles, es decir, las diferentes categorías presentes en una variable.
- Los niveles pueden estar codificados tanto en valores numéricos como caracteres.




```
> starwars
# A tibble: 87 × 14
   name   height  mass hair_color skin_color eye_color birth_year sex   gender homeworld species
   <chr>   <int> <dbl> <chr>      <chr>      <chr>          <dbl> <chr> <chr>  <chr>     <chr>
 1 Luke …    172    77 blond      fair       blue            19   male  mascu… Tatooine  Human
 2 C-3PO     167    75 NA         gold       yellow         112   none  mascu… Tatooine  Droid
 3 R2-D2      96    32 NA         white, bl… red             33   none  mascu… Naboo     Droid
 4 Darth…    202   136 none       white      yellow          41.9 male  mascu… Tatooine  Human
 5 Leia …    150    49 brown      light      brown           19   fema… femin… Alderaan  Human
 6 Owen …    178   120 brown, gr… light      blue            52   male  mascu… Tatooine  Human
 7 Beru …    165    75 brown      light      blue            47   fema… femin… Tatooine  Human
 8 R5-D4      97    32 NA         white, red red             NA   none  mascu… Tatooine  Droid
 9 Biggs…    183    84 black      light      brown           24   male  mascu… Tatooine  Human
10 Obi-W…    182    77 auburn, w… fair       blue-gray       57   male  mascu… Stewjon   Human
# ℹ 77 more rows
# ℹ 3 more variables: films <list>, vehicles <list>, starships <list>
# ℹ Use `print(n = ...)` to see more rows
```




```
starwars %>%
  select(height, mass, sex, homeworld)
```

```
# A tibble: 87 × 4
   height  mass sex    homeworld
    <int> <dbl> <chr>  <chr>
 1    172    77 male   Tatooine
 2    167    75 none   Tatooine
 3     96    32 none   Naboo
 4    202   136 male   Tatooine
 5    150    49 female Alderaan
 6    178   120 male   Tatooine
 7    165    75 female Tatooine
 8     97    32 none   Tatooine
 9    183    84 male   Tatooine
10    182    77 male   Stewjon
# ℹ 77 more rows
# ℹ Use `print(n = ...)` to see more rows
```

```
starwars %>%
  select(height, mass, sex, homeworld) %>%
  summary()
```

```
     height           mass               sex           homeworld
 Min.   : 66.0   Min.   :  15.00   Length   :87   Length   :87
 1st Qu.:167.0   1st Qu.:  55.60   N.unique : 4   N.unique :48
 Median :180.0   Median :  79.00   N.blank  : 0   N.blank  : 0
 Mean   :174.6   Mean   :  97.31   Min.nchar: 4   Min.nchar: 4
 3rd Qu.:191.0   3rd Qu.:  84.50   Max.nchar:14   Max.nchar:14
 Max.   :264.0   Max.   :1358.00   NAs      : 4   NAs      :10
 NAs    :6       NAs    :28
```




```
unique(starwars$sex)
```

```
[1] "male"           "none"           "female"         "hermaphroditic" NA
```

```
starwars %>% distinct(sex)
```

```
# A tibble: 5 × 1
  sex
  <chr>
1 male
2 none
3 female
4 hermaphroditic
5 NA
```

Factores

Vector o variable a categorizar

Etiquetas que toma la variable

```
factor(variable, levels, labels)
```

Niveles que toma la variable

```
sw <-
  starwars %>%
  mutate(
    sexo = factor(sex,
                  levels = c("male", "female",
                             "hermaphroditic", "none"))) %>%
  select(height, mass, sex, sexo, homeworld) %>%
  print()
```

```
# A tibble: 87 × 5
   height  mass sex    sexo   homeworld
    <int> <dbl> <chr>  <fct>  <chr>
 1    172    77 male   male   Tatooine
 2    167    75 none   none   Tatooine
 3     96    32 none   none   Naboo
 4    202   136 male   male   Tatooine
 5    150    49 female female Alderaan
 6    178   120 male   male   Tatooine
 7    165    75 female female Tatooine
 8     97    32 none   none   Tatooine
 9    183    84 male   male   Tatooine
10    182    77 male   male   Stewjon
# ℹ 77 more rows
# ℹ Use `print(n = ...)` to see more rows
```




```
> unique(sw$homeworld)
 [1] "Tatooine"      "Naboo"          "Alderaan"      "Stewjon"       "Eriadu"        "Kashyyyk"
 [7] "Corellia"      "Rodia"          "Nal Hutta"     "Bestine IV"    NA              "Kamino"
[13] "Trandosha"     "Socorro"        "Bespin"        "Mon Cala"      "Chandrila"     "Endor"
[19] "Sullust"       "Cato Neimoidia" "Coruscant"     "Toydaria"      "Malastare"     "Dathomir"
[25] "Ryloth"        "Aleen Minor"    "Vulpter"       "Troiken"       "Tund"          "Haruun Kal"
[31] "Cerea"         "Glee Anselm"    "Iridonia"      "Iktotch"       "Quermia"       "Dorin"
[37] "Champala"      "Geonosis"       "Mirial"        "Serenno"       "Concord Dawn"  "Zolan"
[43] "Ojom"          "Skako"          "Muunilinst"    "Shili"         "Kalee"         "Umbara"
[49] "Utapau"
```

```r
sw <-
  sw %>%
  mutate(planeta = factor(homeworld)) %>%
  print()
```

```
# A tibble: 87 × 6
   height  mass sex    sexo   homeworld planeta
    <int> <dbl> <chr>  <fct>  <chr>     <fct>
 1    172    77 male   male   Tatooine  Tatooine
 2    167    75 none   none   Tatooine  Tatooine
 3     96    32 none   none   Naboo     Naboo
 4    202   136 male   male   Tatooine  Tatooine
 5    150    49 female female Alderaan  Alderaan
 6    178   120 male   male   Tatooine  Tatooine
 7    165    75 female female Tatooine  Tatooine
 8     97    32 none   none   Tatooine  Tatooine
 9    183    84 male   male   Tatooine  Tatooine
10    182    77 male   male   Stewjon   Stewjon
# ℹ 77 more rows
# ℹ Use `print(n = ...)` to see more rows
```




```
> summary(sw)
     height           mass                sex                 sexo        homeworld       planeta
 Min.   : 66.0   Min.   :  15.00   Length   :87   male          :60   Length   :87   Naboo    :11
 1st Qu.:167.0   1st Qu.:  55.60   N.unique : 4   female        :16   N.unique :48   Tatooine :10
 Median :180.0   Median :  79.00   N.blank  : 0   hermaphroditic: 1   N.blank  : 0   Alderaan : 3
 Mean   :174.6   Mean   :  97.31   Min.nchar: 4   none          : 6   Min.nchar: 4   Coruscant: 3
 3rd Qu.:191.0   3rd Qu.:  84.50   Max.nchar:14   NAs           : 4   Max.nchar:14   Kamino   : 3
 Max.   :264.0   Max.   :1358.00   NAs      : 4                       NAs      :10   (Other)  :47
 NAs    :6       NAs    :28                                                        NAs      :10
```



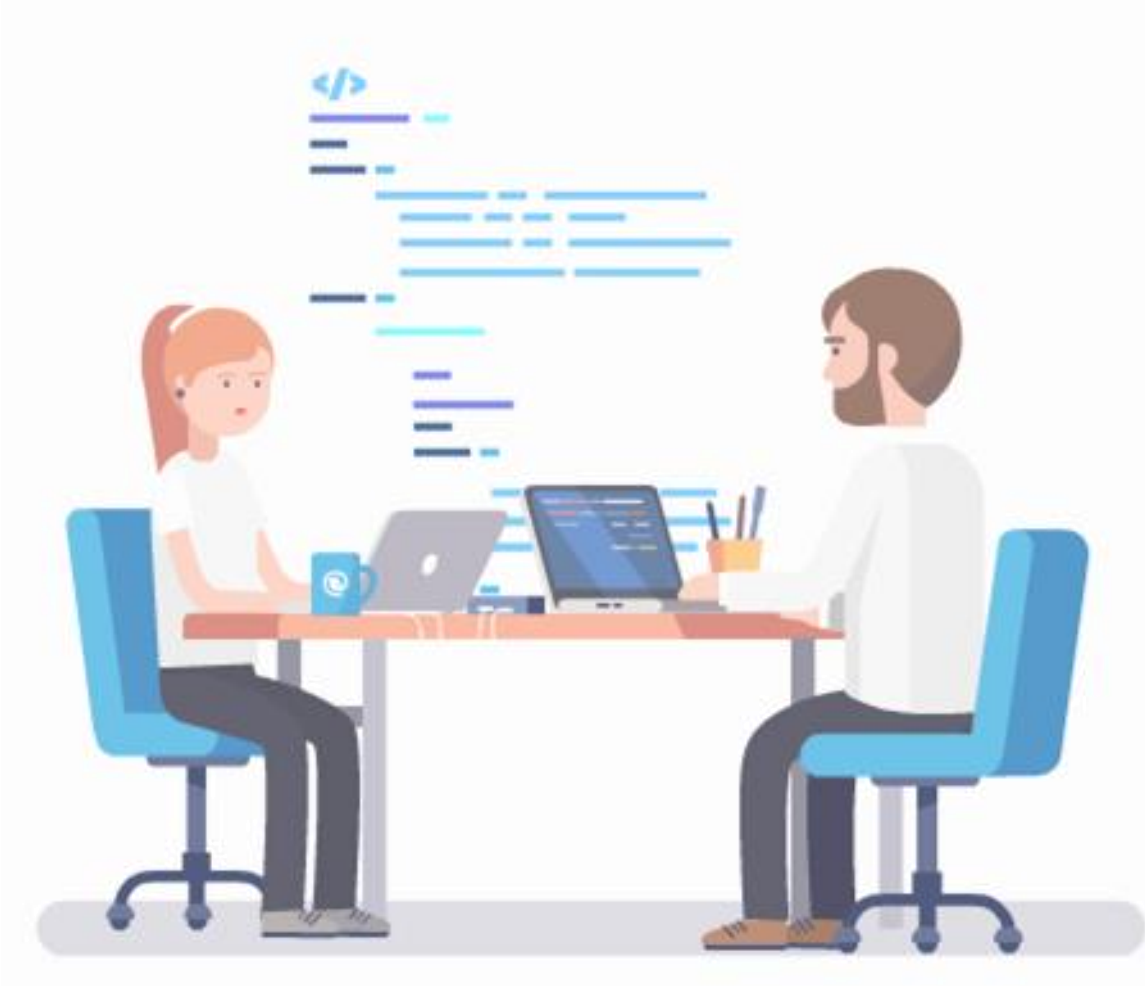
Trabajando con factores en R {forcats}

- Reordenar niveles en malla de datos y/o gráficos
- Rectificar niveles
- Agregar niveles



Su turno...

- Importe la malla promedios_2021_ps.csv
- Nombre a su objeto como **contam**
- Columnas en formato **snake case**






```r
contam <-
  read_csv("./data/promedios_2021_ps.csv", skip = 8) %>%
  clean_names() %>%
  print()
```

```
Rows: 6692 Columns: 5
── Column specification ────────────────────────────────────────────────────────
Delimiter: ","
chr (3): date, id_station, id_parameter
dbl (2): value, unit

ℹ Use `spec()` to retrieve the full column specification for this data.
ℹ Specify the column types or set `show_col_types = FALSE` to quiet this message.
# A tibble: 6,692 × 5
   date       id_station id_parameter value  unit
   <chr>      <chr>      <chr>        <dbl> <dbl>
 1 01/01/2021 ACO        PM10            70     2
 2 01/01/2021 ATI        PM10            44     2
 3 01/01/2021 CUT        PM10            73     2
 4 01/01/2021 FAC        PM10            43     2
 5 01/01/2021 MER        PM10            56     2
 6 01/01/2021 MER        PM2.5           40     2
 7 01/01/2021 NEZ        PM2.5           29     2
 8 01/01/2021 PED        PM10            34     2
 9 01/01/2021 PED        PM2.5           25     2
10 01/01/2021 SAG        PM10            59     2
# ℹ 6,682 more rows
# ℹ Use `print(n = ...)` to see more rows
```

Su turno...

```
summary(contam)
```

```
     date            id_station        id_parameter          value              unit
Length:6692        Length:6692        Length:6692        Min.   :  2.00   Min.   :2
Class :character   Class :character   Class :character   1st Qu.: 17.00   1st Qu.:2
Mode  :character   Mode  :character   Mode  :character   Median : 25.00   Median :2
                                                         Mean   : 30.46   Mean   :2
                                                         3rd Qu.: 39.00   3rd Qu.:2
                                                         Max.   :158.00   Max.   :2
```

- Identifique en la malla contam las variables que requieran codificarse como factor
- Convierta a factor las variables identificadas.
- Nombre a las nuevas columnas como site y pollutant.
- Ejecute el comando summary.
- Elimine las columnas id_station, id_parameter y unit.
- Renombre a la columna value como concentration.